

WEEK 1 · SESSION 1

Welcome to Robotics

Meet your XRP

This month, your code becomes a real robot's behavior.

Today's finish line: **assembled robot + first program running**

BEFORE WE BUILD ANYTHING

What *Is* a Robot?

Not "what parts does it have" — what makes something a robot at all?

Robot or Not?

For each one — robot, or not a robot? Be ready to defend your answer.



Robot vacuum



Dishwasher



RC car (human
driving)



Thermostat



Wind-up toy car



Smart speaker



Vending machine



Mars rover

So, What Makes Something a Robot?

A robot is a system that **senses** its environment, **decides** what to do, and **acts**
— over and over, in a loop.

No sensing, no loop → not a robot, no matter how clever the mechanism.



SHAKY



Watch on  **YouTube**

Every robot does three things

1 · SENSE

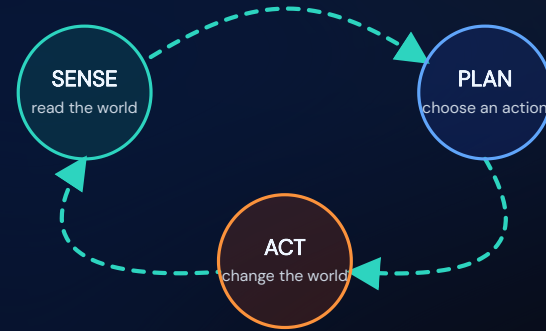
Gather information about the world.

2 · PLAN

Choose what to do with that information.

3 · ACT

Do something that changes the world.



Then the loop repeats—many times per second.

Example: avoiding a wall

SENSE

10 cm

Ultrasonic sensor measures the wall ahead.

PLAN

Too close

The program chooses a turn instead of forward motion.

ACT

Turn away

The motors change the robot's path.

A fresh distance reading starts the next cycle immediately.

Meet the XRP

MOTION

Two independently driven wheels steer without a steering linkage.

BRAIN

A tiny controller runs your MicroPython program.

SENSES

Distance, heading, wheel motion, and floor reflectance.

CONNECTION

A browser-based editor sends code and shows live output.

Small rover. Same building blocks as much larger robots.

Five subsystems make the loop possible

PERCEPTION

Sensors gather data—the “sense” part.

COMPUTATION

Code + processor decide—the “plan” part.

ACTUATION

Motors create motion—the “act” part.

COMMUNICATION

Moves code, data, and messages between devices.

POWER

Supplies the energy every other subsystem needs.

Spot the subsystems on the XRP

Perception

ultrasonic · IMU · reflectance · encoders

Communication

built-in wireless connection to the editor

Computation

RP2040 / RP2350 controller

Power

onboard battery pack

Actuation

two drive motors · servo

Today's lab: bring it to life



UNBOX

Inventory parts and learn their names.



ASSEMBLE

Build the chassis—no soldering required.



CODE

Blink the LED and print one live sensor value.

Ask early. Setup problems are part of the lab, not a test of prior knowledge.

One loop. Five subsystems. Your robot.

REMEMBER

Sense → **Plan** → **Act** repeats continuously.

FIND

Perception · computation · actuation ·
communication · power.

Now let's build an XRP.